

Savitzky-Golay split command-lag search

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rotor median publish period [s]: 0.00495767593384
gimbal median publish period [s]: 0.00495719909668
initial lags [s]: {'rotor': 0.004957675933837891, 'gimbal': 0.0049571990966796875}
smooth half-width multipliers: [4.0, 2.0, 1.0, 0.5]

smooth width=4: rotor=0.0529975335s gimbal=0.00488969025s cost=583.272405792
lag_gradient={'rotor': -2.0942389390543603e-05, 'gimbal': 1.4582688336206573e-05}
FD=[{'direction': 'negative_gradient', 'scheme': 'central', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 29.07731943745076, 'finite_difference_l2_norm': 29.07731943364625,
'difference_l2_norm': 1.3093130493368065e-07, 'relative_error': 4.5028670959622515e-09},
{'direction': 'rotor_delay_seconds', 'scheme': 'central', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 3.6616514933610502, 'finite_difference_l2_norm': 3.6616514171761256,
'difference_l2_norm': 2.412817738088313e-07, 'relative_error': 6.589424860511709e-08},
{'direction': 'gimbal_delay_seconds', 'scheme': 'central', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 6.627175769306424, 'finite_difference_l2_norm': 6.627175669884257,
'difference_l2_norm': 3.5348984366287e-07, 'relative_error': 5.33394399013903e-08}]

smooth width=2: rotor=0.0518558153s gimbal=0.000526932578s cost=583.256588372
lag_gradient={'rotor': 0.0034278420499146423, 'gimbal': 0.045828488010271284}
FD=[{'direction': 'negative_gradient', 'scheme': 'central', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 8.020231757806005, 'finite_difference_l2_norm': 8.020230939571631,
'difference_l2_norm': 2.647764833395536e-06, 'relative_error': 3.3013570097429074e-07},
{'direction': 'rotor_delay_seconds', 'scheme': 'central', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 4.5392563764851905, 'finite_difference_l2_norm': 4.539255716620335,
'difference_l2_norm': 1.8452380659613136e-06, 'relative_error': 4.0650668587926445e-07},
{'direction': 'gimbal_delay_seconds', 'scheme': 'central', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 7.870214888786664, 'finite_difference_l2_norm': 7.870214050355167,
'difference_l2_norm': 2.670395410633823e-06, 'relative_error': 3.3930400228824154e-07}]

smooth width=1: rotor=0.0526728755s gimbal=0.00449300467s cost=583.251946599
lag_gradient={'rotor': -0.00017507741530331522, 'gimbal': 1.5946984072590098e-05}
FD=[{'direction': 'negative_gradient', 'scheme': 'central', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 28.059049166852816, 'finite_difference_l2_norm': 28.05904840178769,
'difference_l2_norm': 6.382782857428373e-06, 'relative_error': 2.2747680505754943e-07},
{'direction': 'rotor_delay_seconds', 'scheme': 'central', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 5.928085135045903, 'finite_difference_l2_norm': 5.928081216042067,
'difference_l2_norm': 6.7144645872146045e-06, 'relative_error': 1.1326531981667656e-06},
{'direction': 'gimbal_delay_seconds', 'scheme': 'central', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 8.866226822998408, 'finite_difference_l2_norm': 8.866223097893748,
'difference_l2_norm': 1.3133694929729451e-05, 'relative_error': 1.4813172719269386e-06}]

smooth width=0.5: rotor=0.051063552s gimbal=0.00418174478s cost=583.245190843
lag_gradient={'rotor': 0.00032619226264785084, 'gimbal': 8.386924355008318e-08}
FD=[{'direction': 'negative_gradient', 'scheme': 'central', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 66.25523089038161, 'finite_difference_l2_norm': 66.2552306339133,
'difference_l2_norm': 3.499978858579023e-06, 'relative_error': 5.282569861343765e-08},
{'direction': 'rotor_delay_seconds', 'scheme': 'central', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 11.579281768693072, 'finite_difference_l2_norm': 11.579242933689395,
'difference_l2_norm': 5.175577273607898e-05, 'relative_error': 4.469687651613347e-06},
{'direction': 'gimbal_delay_seconds', 'scheme': 'central', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 9.363084369854425, 'finite_difference_l2_norm': 9.363062583754697,
'difference_l2_norm': 7.944450766831537e-05, 'relative_error': 8.484865086135131e-06}]

strict profile iterations: 1
iteration 1: best=(0.051063551966, 0.004181744776) cost=583.242163243
rotor_range=(0.046105876032, 0.0560212279) gimbal_range=(0.0, 0.009138943872) expansions=[]
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selected rotor lag 0.051063551966 s (10.2999 publish periods)
selected gimbal lag 0.004181744776 s (0.84357 publish periods)
selected objective cost 583.242163028